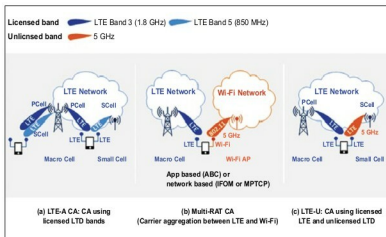


Will LTE-U destroy Wi-Fi



Comparison between LTE-A CA, Multi-RAT CA and LTE-U

There seems to be a battle going on between the proponents of Wi-Fi and those keen to introduce a new technology—LTE in Unlicensed Spectrum (LTE-U). Besides the expensive spectrum that companies like Airtel, Reliance and Vodafone bid for, there are unlicensed airwaves that are traditionally used for cordless phones, remote controls and monitors, etc. Wi-Fi also makes use of this free spectrum. But companies like Verizon and Qualcomm are getting ready to exploit this unlicensed spectrum for what

they claim would be a better mobile broadband experience.

The proposal for LTE-U was originally given by Qualcomm for using 4G LTE radio communications technology in unlicensed spectrum such as 5GHz band, which is used by dual-band Wi-Fi equipment. Verizon is pushing to make it popular by equipping phones with chips that can make better use of LTE-U.

LTE-U will not replace LTE; instead it will improve LTE's speed and coverage in areas having heavy congestion. It is a small-cell technology unlike LTE that is broadcast from large towers. What's good is that, the devices currently making use of Wi-Fi will be able to run on LTE-U as well. But the performance of Wi-Fi can suffer, in case of bad network architecture, in the presence of LTE-U.

However, there is a possibility that Wi-Fi and LTE-U may co-exist after proper mechanisms have been arrived at.

Based on 3GPP Rel. 12, LTE-U targets early mobile operators' deployments in the USA, South Korea and India, with co-existence tests defined by LTE-U Forum. Likewise, defined in 3GPP Rel. 13, Licensed Assisted Access (LAA) targets deployments in Europe, Japan and some other countries. LTE-Wi-Fi Link Aggregation (LWA) will be used for deployments where the carriers already have Wi-Fi in use.

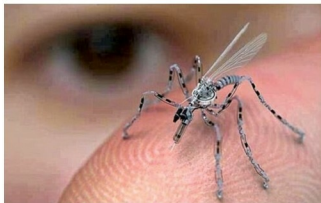
A microdrone, the spying tool of the future

A team of researchers at Johns Hopkins University, USA, in conjunction with US Air Force Office of Scientific Research at Wright-Patterson Air Force Base in Arlington, Virginia, is helping develop what they are calling a micro aerial vehicle (MAV) that will undertake various espionage tasks.

The robotic insect can effortlessly infiltrate urban areas, where dense concentrations of buildings and people, along with unpredictable winds and other obstacles make it impractical. It can also land precisely on human skin, use its super-micron size needle to take DNA samples and fly off again at speed. All people feel is the pain of a mosquito bite without the burning sensation and the swelling.

The device can inject a micro radio frequency identification (RFID) tracking device right under the skin and can be used to inject toxins into enemies during wars.

The device can be controlled from a great distance and is equipped with a camera and built-in microphone.



US Air Force's micro aerial vehicle